TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

Proposer	Name	Dr. Loïc Burr	Email	loic.burr@csem.ch
Institute Name	CSEM			
Address	Rue Jaquet-Droz 1, 2002 Neuchâtel, Switzerland			

2. Technology Overview

Title of Proposed Technology	**BioTracker** **A Modular Salivary Biosensing Platform for Multi-Analyte, Non-Invasive Health Monitoring**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (X), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Bio-signal Measurement / Diagnostic Devices	Medical Information & Systems	Remote and Home Health care Devices
Concept of Proposed Technology	The proposed BioTracker technology is a modular, non-invasive salivary biosensing platform designed to deliver rapid, quantitative, multi-analyte health insights from a single saliva sample. The system includes: - Disposable microfluidic cartridge - with on-cartridge sample preparation (dilution, conditioning), integrated electrochemical sensor array, and fluidic controls designed for saliva's viscosity/foaming behavior. - Reusable electronic reader - with low-noise analog front-end, temperature and reference control, and Bluetooth/USB connectivity to mobile/PC apps for guided testing and results reporting. - Embedded analytics - for on-cartridge calibration strategies and cross-analyte compensation; cloud/API hooks for data aggregation and AI-enabled insights (optional). This architecture enables simultaneous measurement of key biomarkers such as ions (e.g., K^+, Na^+), metabolites (e.g., lactate, glucose), and selected proteins or hormones, making it adaptable to multiple health-related panels. Its core functions include fully guided sample handling, automated on-chip analysis, and real-time result interpretation.			
Rationale for Joint R&D Project	The BioTracker platform addresses a broad range of high-value applications in Korea's expanding digital-health, oral-care, sports-performance, and preventive-care markets. As a multi-analyte salivary diagnostic system, it supports frequent, non-invasive monitoring and can be **tailored to both consumer and clinical** workflows. **Potential Uses** - Routine self-monitoring of electrolyte balance, hydration, and daily metabolic trends (e.g., lactate, glucose) for fitness and lifestyle management. - Screening of inflammation-related biomarkers to support early detection of oral or systemic			

	conditions. - Oral-care applications, including markers of gum disease and oral microbiome activity. - Sports performance monitoring, offering repeatable, non-invasive measurements before, during, and after training. - Telemedicine and home-healthcare, enabling regular check-ups for elderly users, chronic-disease management, and wellness subscription services. - Modular architecture enabling the SME to create multiple product lines—from entry-level wellness tools to clinically oriented diagnostics—based on the same core system. **Ideal Korean SME Partner** The ideal Korean SME will drive the application-focused development of the BioTracker platform by defining the key use-cases, shaping product requirements, and ensuring thesolution fits Korean consumer, wellness, and healthcare markets. As an innovation-drive company active in digital health, medical devices, oral-care technology, AI-enabled analytics, or consumer wellness hardware, the SME will translate the core technology into compelling, user-relevant products and service offerings. The Korean SME partner shall take responsibility of the medical certification process and arrange clinical studies.
Key Words	- Non-invasive salivary point of care diagnostics - Decentralized health - Electrochemical biosensors - Microfluidic cartridge - Portable electrochemical reader

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 4	End of the Project	Level 6
R&D Objectives	The primary objective of this joint R&D project is to advance the BioTracker platform from a validated laboratory concept to a robust, manufacturable, multi-analyte salivary diagnostic solution (TRL4→TRL6) usable for clinical demonstration. To achieve this, the project will develop and refine several key technological building blocks: 1. **a next-generation disposable microfluidic cartridge**, engineered for saliva collection, on-chip calibration solution storage and sample conditioning, controlled fluid transport, and stable multiplexed sensor interfacing 2. **a modular electrochemical multiplexed sensor array**, covering analytes of interest for the use case, including ions, metabolites, and at least one protein or hormone, supported by optimized surface functionalization and assay chemistry 3. **a compact, low-noise electronic reader**, integrating signal acquisition, cartridge actuation, temperature and reference-electrode control, and secure data transmission 4. **embedded analytics and AI-assisted algorithms**, enabling multi-analyte fusion, drift correction, and contextual interpretation 5. **design for manufacturability / regulatory pathway**, ensuring that each subsystem - cartridge, sensors, electronics, and software - can be industrialized and commercialized by the Korean SME partner. These developments collectively create an integrated platform that combines microfluidics, electrochemical biosensing, and embedded intelligence to deliver reliable, real-time health insights from a single saliva sample.			
R&D Target Specifications	The BioTracker platform aims to deliver a robust multi-analyte salivary diagnostic system with the following performance targets: - **Analytical performance:** simultaneous quantification of ≥4 analytes with ±10% accuracy. - **Sample handling:** ≤150 μL saliva input, full workflow completed in ≤15 minutes, with integrated on-chip- dilution (1:1–1:10) and bubble suppression. - **Cartridge performance:** injection-moldable, <USD 5, ≥12month shelf- -life, >90% electrode yield. - **Sensor/reader performance:** 4–6 electrochemical channels, reader sensitivity ≤1 μV (OCP) and ≤10 pA (current), Wifi/BLE/USB-C connectivity, and temperature control ±1°C. - **Software & analytics:** real-time filtering, automatic calibration, cloud integration and encrypted data transfer, and AI-based matrix-effect correction reducing error by ≥20%. - **System level readiness:** fully integrated TRL4→TRL6 prototype, ≤3 user steps, <5% test failure rate, and production readiness of prototype devices and cartridges for Korean pilot deployment.			

R&D Plan	**Phase 1: System Definition (Korean SME) & Core Architecture (CSEM)** - Define priority use-cases with the Korean SME and establish market-driven requirements. - Freeze system specifications for cartridge, sensor array, reader, and analytics pipeline. - Conduct saliva-specific feasibility studies (sample intake, dilution, foaming/bubbles, matrix variability). - Select target analytes (ions, metabolites, 1 protein/hormone) and select benchmark methods. - Deliverables: System Requirements, Architecture Specification, Verification & Risk Plan. **Phase 2: Technology Development & Component Prototyping (CSEM)** - Develop microfluidic cartridge: saliva intake geometry, on-chip dilution/buffering, controlled wetting, sensor interfacing. - Engineer modular electrochemical sensors: ISE (K^+/Na^+), enzymatic (lactate/glucose), protein/hormone channel; drift & selectivity optimization. - Build reader electronics: low-noise multi-channel AFE, temperature/reference control, Wifi/BLE/USB-C connectivity, guided HMI. - Implement embedded analytics: calibration algorithms, drift compensation, multi-analyte fusion logic. - Deliverables: Alpha/Beta cartridge versions; EVT/DVT reader prototypes; analytics v1.0. **Phase 3: System Integration (CSEM), Verification & Pilot Preparation (Korean SME)** - Integrate cartridge, sensors, reader, firmware, and analytics into a functional TRL-5 system. - Perform verification studies: accuracy/precision vs. benchtop, matrix-effect robustness, drift stability, usability tests (≤15-min workflow). - Manufacture pre-series pilot batch. - Prepare pilot: protocols, KPI definitions, test site coordination, preliminary regulatory (K-FDA) pathway. - Deliverables: Verified integrated system, pre-series hardware, pilot readiness package. **Phase 4: Pilot Deployment, Optimization & Industrialization (Korean SME)** - Field pilot: evaluate performance, usability, device reliability, and failure rate. - Optimize cartridge, sensors, reader, and analytics based on real-world feedback. - Localize supply chain with the Korean SME and prepare training + tech-transfer documentation. - Regulatory preparation and filling - Deliverables: TRL-6 prototype platform, industrialization package, commercialization roadmap for the Korean market.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	CSEM will lead the development, integration, and validation of the enabling technologies forming the BioTracker platform, focusing on robust, modular components that can be directly transferred into the SME's product development and manufacturing workflows. - Develop the saliva microfluidic cartridge, electrochemical sensor array, reader electronics, and embedded analytics/AI. - Conduct core research, characterization, and verification, ensuring stable multi-analyte performance and system reliability. - Provide design files, SOPs, documentation, and structured technology transfer, enabling industrialization and scale-up. - Support design for manufacturing readiness, and early regulatory documentation. - Supply advanced R&D infrastructure and multidisciplinary expertise (biosensing, microfluidics, electronics, data science).
Expected R&R of Korean SME Partner	The Korean SME will drive the application-focused development and commercialization strategy, steering product requirements, market fit, and adoption to ensure that BioTracker evolves into a competitive solution for the Korean health and lifestyle market. - Define use-cases, user requirements, and product priorities, ensuring alignment with Korean customer expectations. - Lead productization including industrial design, UX, mobile/cloud integration, and service-model development. - Take key responsibility for manufacturing co-design, cost optimization, supplier selection, and production scaling. - Conduct testing, validation, and pilot deployment in Korea, collecting real-world performance and usability data. - Lead regulatory preparation, market-entry planning, and commercialization activities. - Commit engineering and operational resources and maintain continuous collaboration with CSEM throughout development and industrialization.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project will deliver a TRL4→TRL6 **multi-analyte salivary diagnostic platform** integrating a validated microfluidic cartridge, biosensor array, compact reader, and embedded analytics. The outcome will be a pilot-ready, industrializable system enabling rapid, non-invasive health insights and strategically positioning the Korean SME in the expanding digital-health and preventive-care markets. Beyond the core technology, the project will generate a scalable product foundation, manufacturing know-how, and market-relevant evidence that support both immediate commercialization and long-term sectoral impact. **Technological Achievements** - Integrated TRL6 BioTracker prototype with validated saliva microfluidics, sensor array, electronics, and analytics. - Demonstrated multi-analyte accuracy and robustness for ions, metabolites, and one protein/hormone. - Verified **saliva-specific robustness** (viscosity, foaming, matrix variability) and reliable sample-to-answer workflow. **Commercialization Potential** - Manufacturing-ready design package enabling localization of pre-series production by the Korean SME. - Pilot-validated performance supporting K-FDA preparation and market introduction. - Platform architecture enabling rapid extension to new biomarkers and product variants. **Broader Industrial & Economic Impact** - Strengthened competitiveness of the Korean SME in digital health, preventive-care, oral-care, and wellness markets. - Establishment of a new non-invasive diagnostic category supporting decentralized, home-based monitoring and early detection. - Development of local supply chains, production knowhow, and high value technical capabilities within Korea. - Foundation for future international expansion, product variants, and data-driven services. **Scientific & Market Evidence Generation** - High-quality performance datasets validating the use of salivary biomarkers for continuous health insight and early-risk detection. - Pilot evidence enabling market differentiation, improved user engagement, and integration into Korean digital-health ecosystems.

Saliva collection cartridge concept

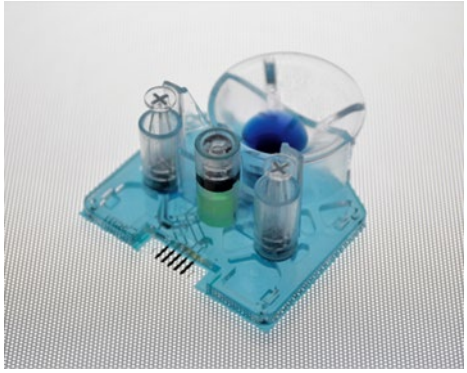

Figure 1: Photography of a single analyte analysis cartridge prototype for saliva collection, pretreatment and electrochemical analysis.

6. References

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization